	7.13 CHARACTERISATION AND ASSIGNMENT OF PROPERTY VALUES AND THEIR UNCERTAINTIES	
	Document no: ASMF 7.13	
Compiled by:	Tshepo Maponya – Laboratory Supervisor	
Approved by:	Sifiso Nyambi- Laboratory Manager	

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1. Purpose

To define the processes used by the lab to:

- Characterise candidate reference materials (RMs),
- Assign certified values and associated uncertainties,
- Ensure compliance with ISO 17034:2016 and SANAS F149 clauses **7.12 and 7.13**,
- Ensure statistical rigor according to ISO 13528 (Algorithm A),
- Ensure suitability of certified coal reference materials for calibration, QC and PT schemes.

2. Scope

- Ash content (% dry basis) – ISO 1171 / Quick Ash
- Calorific Value (MJ/kg, dry basis) – ISO 1928
- Moisture (MIAS) – SANS 5925
- Volatile Matter (% dry basis) – ISO 562
- Total Sulphur (% dry basis) – ASTM D4239
- Ash Fusion Temperature (Reducing) – ISO 540
- Hardgrove Grindability Index – ISO 5074

This procedure applies to all CRM batches produced and intended for certification.

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3. Reference Documents

- ISO 17034:2016
- ISO 17043:2023
- ISO 33405:2024
- ISO/IEC 17025:2017
- ISO 5725 (Parts 1–6): Accuracy of measurement methods
- ASTM/ISO coal testing methods (e.g., ISO 1171, ISO 562, etc.)

4. Responsibilities

- Analyst: Conducts characterisation measurements
- Laboratory Manager/Supervisor: Process data, calculate assigned values, reviews and approves characterisation reports
- Operations Manager: Reviews and approves all procedures, technical forms & CRM certificates.

5. Definitions

- **CRM** – Certified Reference Material (ISO 17034).
- **Characterisation** – Determination of property values using scientific measurement processes.
- **Certified value** – Property value assigned to a CRM with stated uncertainty and traceability.
- **Homogeneity / Stability** – As defined in ISO 17034 & ISO 13528.
- **Algorithm A** – Robust consensus estimation procedure per ISO 13528.
- **K=2** – Coverage factor for approximate 95% expanded uncertainty.

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6. Metrological Traceability

All certified values shall be traceable to:

Where SI traceability is possible (CV &TS):

- Benzoic acid pills for calorimetry
- Balances and thermometers calibrated through SANAS accredited labs, traceable to NMI.

Where traceability Is not possible(Proximates, AFT, HGI)

- Traceability is endured through:
- Use of validated reference measurement procedures(ISO 1171, ISO 562, SANS 5925, ISO 540, ISO 5074)
- Use of consensus PT assigned values from ISO 13528 Algorithm A.
- Use of competent CRM's where available
- Document chain of calibrations for balances, ovens, furnaces, thermocouples.
- Validate methods derived from internationally recognized standards(ISO & ASTM)

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7. Characterization Requirements (ISO 17034 clause 7.12)

7.1. **Characterisation Strategy**

Azuresky uses two characterisation pathways:

1. **Single reference measurement procedure in a single competent laboratory (ISO methods)** – Clause 7.12.3(a)
2. **Characterisation using results from interlaboratory comparison (PT) using ISO 13528 Algorithm A** – Clause 7.12.3(c)

These strategies are appropriate because:

- All CRM properties are quantitative.
- All measurands are operationally defined by international coal standards.

The laboratory has demonstrated technical competence through ISO/IEC 17025-accredited methods.

7.2. Definition of Measurands

Property	Measurand Definition	Method
Ash	% dry basis at 815°C	ISO 1171 / Quick Ash
CV	MJ/kg dry basis	ISO 1928
MIAS	% moisture	SANS 5925
VM	% dry basis	ISO 562
Total S	% dry basis	ASTM D4239

7.3. Measurement Plan

A detailed measurement plan shall be prepared for each CRM batch including:

- Number of replicates per property (minimum 10 bottles).
- Equipment and method to be used.
- Single-analyst requirement for each property.
- Duration: characterisation performed consecutively or over several days if needed.
- Environmental conditions (documented).

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- Calibration validity checks prior to measurements.
- Data recording format and integrity controls.

The plan must be authorised by the Lab Manager and communicated to personnel

7.4. Execution of characterization

- Pulverised material is homogenised (4 hours) and sub-sampled using flat & heap into 150 g bottles.
- 10 bottles are analysed for each parameter using the designated ISO standard.
- One analyst performs the measurements, using validated ISO/IEC 17025 methods.
- All measurements are repeated if quality control samples fall outside control limits.

7.5. Outlier Testing

Outliers shall be evaluated using the **Grubbs test** (ISO 5725).
Outliers may only be removed after:

- Investigating analytical cause,
- Documenting justification,

Approval by the Lab Manager

7.6 Technical Evaluation

The Lab Manager performs:

- Verification of adherence to the measurement plan.
- Review of raw data, calibration data, QC results.

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- Confirmation of method performance (accuracy, repeatability, instrument stability).
- Investigation of any deviations.

If deviations are significant, data are excluded and measurements repeated.

7.7. Control Environmental Conditions

Ensure environmental conditions (e.g. temperature, humidity) are suitable for accurate and stable measurements.

7.8. Statistical Treatment

- Use means or robust statistical estimators (e.g. robust mean, median).
- Identify and eliminate outliers using ISO 5725, Dixon's Q, or Grubbs' test or any other sound statistical means.
- Calculate the standard deviation of the mean or robust uncertainty.

5.8. Assess Method Uncertainty

- Include method repeatability, bias, calibration, and sample preparation effects.
- Uncertainty should be documented in an uncertainty budget.

5.9. Assign Property Values

- The assigned value is the best estimate of the true value based on valid statistical analysis.
- Include expanded uncertainty (U = k × u), coverage factor (usually k=2 for 95% confidence).
- Property values and their uncertainties to be concluded after characterization has been completed.

5.10. Peer Review / Technical Review

- Independent review by technical staff not involved in measurement.

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- Check method validation, traceability, uncertainty, and data integrity.

8. Assignment of Property Values (ISO 17034 Clause 7.13)

8.1. **Statistical Method**

Certified values are assigned using **ISO 13528 Algorithm A (robust mean)** from:

- Consensus PT results from competent coal laboratories.
- Internal characterisation results as supporting evidence.

Algorithm A provides:

- Robust mean (x*),
- Robust standard deviation (s*),

Resistance to outliers and non-normal data

8.2. Experimental Design Requirements

The experimental design requirements follow that of the developed PT scheme given that the CRMs are a derivative of the PT scheme. As such, the experimental design for CRM will be taken from the experimental design of the PT and the extension to CRM will be the evaluation of all parameters by the lab for homogeneity and stability to account for this later in value assignment and uncertainty evaluation, together with the uncertainty of the PT. Transport is a negligible factor since coal is not a highly volatile substance. This is covered in ASMF 7.2.2 Statistical Design.

The design simply put, will follow this structure:

- Characterisation measurements (10 units).
- Homogeneity ANOVA (SS_between vs SS_within).
- Stability assessment (initial and monthly).
- PT consensus computation using Algorithm A.

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- Combination of uncertainty components.

8.3. Outliers Policy

As per ISO 17034 and SANAS F149:

- Outliers are not removed automatically.
- Grubbs test guides investigation.
- Removal allowed only with technical justification (anything outside ±3 z-score guard band is treated as an outlier).

Robust statistics (Algorithm A) minimise need for removal as an alternative approach.

8.4. Weighting Methods

Since results come from equivalent ISO methods, no differential weighting is applied. All PT participant results carry equal weight under ISO 13528.

8.5. Uncertainty Evaluation

Final certified uncertainty includes:

1. **Characterisation uncertainty (u_char)**
   Standard deviation of characterisation mean / √n
2. **Homogeneity uncertainty (u_hom)**
   From ANOVA per ISO 13528 Annex B
3. **Stability uncertainty (u_stab)**
   From initial monitoring
4. **Calibration & instrumental uncertainty (u_cal)**
   From balance, oven, temperature, crucibles, etc.

Combined uncertainty:

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$$u_c = \sqrt{u_{char}^2 + u_{hom}^2 + u_{stab}^2 + u_{cal}^2}$$

Expanded uncertainty:

$$U = k \times u_c(k = 2)$$

8.6. Assignment of Certified Value

Certified value is issued as:

Certified V alue $= x^* \pm U$

Where x* is the ISO 13528 robust mean (Algorithm A).

This value is used in CRM certificates and PT reports

9. Release of CRM Certificates

CRM certificates shall include (ISO 17034 & F149 7.14):

- Title & unique identifier
- Batch number
- Intended use
- Storage conditions
- Property values + uncertainty (k=2)
- Traceability statements
- Method references
- Validity period

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- Handling instructions
- Signature & role of approving authority
- Commutability statement where applicable

10. Documentation and Reporting

The following should be included in the characterisation report:

- Measurement plan
- Raw data from all characterisations runs
- Outlier investigation records
- Homogeneity ANOVA outputs
- Stability monitoring logs (12 months)
- Instrument calibration certificates
- Uncertainty calculations
- Final CRM Certificate
- Personnel involved
- Version-controlled approval record

The records should be archived for a period of 5 years.

11. Controls and Records

Record	Description
Raw data sheets	Original measurement results
Calibration certificates	For all measurement instruments used
Uncertainty budget	Used to estimate uncertainty
Statistical reports	All analysis used to determine certified values

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Characterisation reports	Final signed report for each batch

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